

Mathematics of Computing

Sets: Set-Builder Form, Types, Operations & Hasse Diagrams

Aishwarya J S

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Outline

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- 2 Types of Sets
- 3 Set Operations
- 4 Inclusion–Exclusion Principle
- 5 Partially Ordered Sets & Hasse Diagrams

Sets and Set-Builder Form (1–3)

$$\begin{aligned} \textcircled{1} \quad S &= \{-5, -3, -1, 1, 3, 5\} \\ S &= \{x \mid x \in \mathbb{Z}, x \text{ is odd}, -5 \leq x \leq 5\} \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad S &= \left\{ \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6} \right\} \\ S &= \left\{ \frac{1}{x} \mid x \in \mathbb{Z}, 2 \leq x \leq 6 \right\} \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad S &= \{1, 3, 6, 10, 15, 21, \dots\} \quad (\text{triangular numbers}) \\ S &= \left\{ x \mid x = \frac{n(n+1)}{2}, n \in \mathbb{N} \right\} \end{aligned}$$

Sets and Set-Builder Form (4–6)

$$\begin{aligned} 4 \quad S &= \{\dots, -10, -5, 0, 5, 10, \dots\} \\ S &= \{x \mid 5 \mid x, x \in \mathbb{Z}\} \end{aligned}$$

$$\begin{aligned} 5 \quad S &= \{1, 5, 9, 13, 17, \dots\} \\ S &= \{x \mid x \equiv 1 \pmod{4}, x \in \mathbb{N}\} \end{aligned}$$

$$\begin{aligned} 6 \quad S &= \left\{1, -1, \frac{1}{2}, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{3}, \dots\right\} \\ S &= \left\{x \mid x = \frac{(-1)^n}{n}, n \in \mathbb{N}\right\} \end{aligned}$$

Sets and Set-Builder Form (7–10)

7 $S = \{0, \pm 1, \pm 4, \pm 9, \pm 16, \dots\}$
 $S = \{x \mid x = (-1)^n n^2, n \in \mathbb{W}\} \quad (\mathbb{W} = \text{whole numbers})$

8 **Mersenne numbers:** $M = \{1, 3, 7, 15, 31, 63, \dots\}$
 $M = \{x \mid x = 2^n - 1, n \in \mathbb{N}\}$

9 $S = \{(1, 9), (2, 8), (3, 7), (4, 6), (5, 5), \dots\}$
 $S = \{(x, y) \mid x + y = 10, (x, y) \in \mathbb{N} \times \mathbb{N}\}$

10 $S = \{(1, 1), (2, 8), (3, 3), (4, 4), (8, 8), \dots\}$
 $S = \{(x, y) \mid (x, y) \in \mathbb{N} \times \mathbb{N}, xy = n^2, n \in \mathbb{N}\}$

Types of Sets: Cardinality

Definition

The **cardinality** of a set is the number of distinct elements in it, denoted $|A|$ or $n(A)$.

Example

$$A = \{2, 4, 6, 8\} \implies |A| = 4$$

$$A = \{a_1, a_2, \dots, a_n\} \implies |A| = n$$

ML/AI Examples

- Feature set: $F = \{\text{Age, BMI, Glucose, Blood Pressure, Insulin}\}$, $|F| = 5$
- NLP vocabulary: $V = \{w_1, w_2, \dots, w_{50000}\}$, $|V| = 50000$

Finite and Infinite Sets

Finite Set

Contains a finite number of elements; there exists $n \in \mathbb{N}_0$ such that $|A| = n$.

Example

Feature set

$F = \{\text{Age, BMI, Glucose, BP, Insulin}\}$, $|F| = 5$

Infinite Set

Cardinality is infinite ($|A| = \infty$).

Examples

Feature space $X = \mathbb{R}^3$ (age, height, weight)
 $\Rightarrow |X| = \infty$

Weight space $W = \mathbb{R}^n$ of a neural network
 $\Rightarrow |W| = \infty$

Null (Empty) Set

Definition

A set that contains no elements. $A = \{\}$ or $\emptyset$, $|A| = 0$.

Examples

ML dataset query:

$$D = \{x \in \text{Training Dataset} \mid \text{Age} < 0\} = \emptyset$$

Computer Vision:

$$C = \{x \in \text{Image Dataset} \mid x \text{ is both a cat and a car}\} = \emptyset$$

Equal Sets

Definition

Two sets are **equal** if they contain exactly the same elements, irrespective of order:

$$A = B \iff \forall x (x \in A \iff x \in B) \iff A \subseteq B \text{ and } B \subseteq A$$

Example: Graph Neural Networks – Edge Set

Suppose a graph is represented as:

$$E_1 = \{(v_1, v_2), (v_2, v_3), (v_3, v_4)\}, \quad E_2 = \{(v_3, v_4), (v_1, v_2), (v_2, v_3)\}$$

Since both sets contain exactly the same edges: $E_1 = E_2$.

Equivalent Sets

Definition

Two sets A and B are **equivalent** if they have the same cardinality, i.e. there exists a bijective function $f : A \rightarrow B$:

$$A \sim B \iff \exists f : A \rightarrow B \text{ such that } f \text{ is one-to-one and onto}$$

Example: Supervised Learning – Training samples and Labels

Let $X = \{x_1, x_2, \dots, x_N\}$ be training samples and $Y = \{y_1, y_2, \dots, y_N\}$ be the corresponding labels.

Define $f : X \rightarrow Y$, $f(x_i) = y_i$.

Since each training sample has exactly one corresponding label: $X \sim Y$.

Subset and Superset

Subset

A is a subset of B if every element of A is also in B :

$$A \subseteq B \iff \forall x \in A \Rightarrow x \in B$$

Superset

A is a superset of B if every element of B is also in A : $A \supseteq B$

$$A \supseteq B \iff \forall x (x \in B \Rightarrow x \in A)$$

Example

Let $D = \{(x_i, y_i)\}_{i=1}^N$ be a complete training dataset, and $T = \{(x_i, y_i)\}_{i=1}^{1000}$ a validation subset. Then $T \subseteq D$.

Example: NLP Vocabulary

$V = \{w_i\}_{i=1}^{50000}$ complete vocabulary.
 $V' = \{w \in V \mid \text{freq}(w) \geq 100\}$ high-frequency words. Since $V' \subseteq V$, also $V \supseteq V'$.

Universal Set

Definition

The set of all elements under consideration in a given context, U . For any set A in that context, $A \subseteq U$.

Example: ML Dataset Universe

Let $U = \{(x_i, y_i)\}_{i=1}^N$ represent the complete labeled dataset of a supervised learning problem. Let $T, V, S \subseteq U$ be the training, validation and test datasets respectively. Then $T \subseteq U$, $V \subseteq U$, $S \subseteq U$.

Power Set

Definition

The power set $\mathcal{P}(A)$ (or 2^A) is the set of *all* subsets of A , including the empty set and A itself.

$$\mathcal{P}(A) = \{x \mid x \subseteq A\}, \quad |A| = n \implies |\mathcal{P}(A)| = 2^n$$

Example: ML Feature Selector

Let $F = \{x_1, x_2, x_3\}$ be the set of input features of an ML model.

$$\mathcal{P}(F) = \{\emptyset, \{x_1\}, \{x_2\}, \{x_3\}, \{x_1, x_2\}, \{x_1, x_3\}, \{x_2, x_3\}, \{x_1, x_2, x_3\}\}$$

Since $|F| = 3$: $|\mathcal{P}(F)| = 2^3 = 8$. Every subset represents a possible combination of input features for training a model.

Singleton Set

Definition

A set that contains exactly one element. $|A| = 1$, $A = \{x\}$.

Example: ML Predicted Class

Consider a classifier model $f : X \rightarrow Y$ where $Y = \{\text{Cat}, \text{Dog}, \text{Car}\}$. For an input image x , suppose $f(x) = \text{Dog}$. The predicted set is $P = \{\text{Dog}\}$. Since $|P| = 1$, P is a singleton set.

Countable Set

Definition

A set whose elements can be put into a one-to-one correspondence with the set of natural numbers $\mathbb{N}$. A countable set may be finite or countably infinite.

Example: NLP Vocabulary

Let $V = \{w_1, w_2, w_3, \dots\}$ be the vocabulary of a language model. Define $f : \mathbb{N} \rightarrow V$, $f(i) = w_i$. Since each word has a unique index, $V \sim \mathbb{N}$, so V is a countable set.

Union of Sets

Definition

The union of two sets A and B is the set of all elements that belong to A , or B , or both.

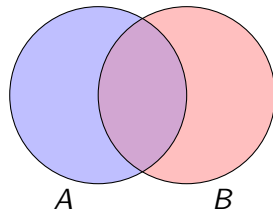
$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

Example: ML Feature Fusion

$F_1 = \{\text{Age, BMI, BP}\}$ (clinical),

$F_2 = \{\text{Glucose, Insulin}\}$ (lab).

$F = F_1 \cup F_2 = \{\text{Age, BMI, BP, Glucose, Insulin}\}$



Intersection of Sets

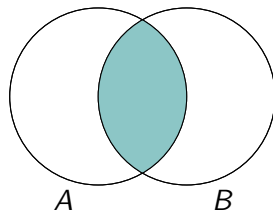
Definition

The intersection of two sets A and B is the set of all elements common to both.

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

Example: Image Dataset Filtering

Let D_1 be the set of images containing cats, D_2 the set of images captured at night. Then $D = D_1 \cap D_2$ = night-time cat images, usable to train/evaluate autonomous driving models.



Set Difference

Definition

The set difference $A - B$ is the set of all elements that belong to A but not to B .

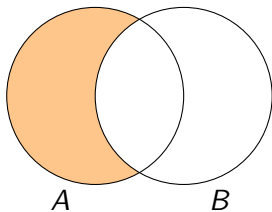
$$A - B = \{x : x \in A \text{ and } x \notin B\}$$

Example: Training Set Construction

Let D be the complete dataset and T the test dataset. The training dataset is

$$\text{Train} = D - T$$

This removes all test samples from the complete dataset before training the model.



Symmetric Difference

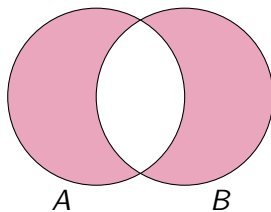
Definition

The symmetric difference of A and B is the set of elements that belong to either A or B , but not to both.

$$A \triangle B = (A - B) \cup (B - A) = (A \cup B) - (A \cap B)$$

Example: Dataset Comparison in CV

Let D_1 be images detected as cars by Model 1, D_2 images detected as cars by Model 2. $D_1 \triangle D_2$ contains images classified as cars by only one of the two models – useful for analyzing disagreements in model performance evaluation.



Complement

Definition

The complement of a set A (denoted A' , A^c , or $U - A$) is the set of all elements in the universal set U that are not in A .

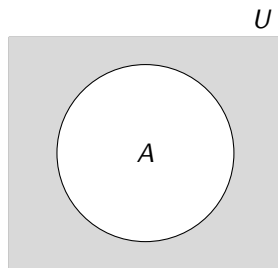
$$A' = \{x : x \notin A \text{ and } x \in U\}$$

Example: Dataset Partitioning

Let $U = \{(x_i, y_i)\}_{i=1}^N$ be the complete labeled dataset, and $T \subseteq U$ the training dataset. The complement of T is

$$T' = U - T$$

which represents all samples *not* included in training (e.g. validation/test samples).



Cartesian Product

Definition

The Cartesian product of two sets A and B is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$.

$$A \times B = \{(x, y) : x \in A \text{ and } y \in B\}, \quad |A| = m, |B| = n \implies |A \times B| = mn$$

Example: Feature Vectors and Class Labels

Let $X = \{x_1, x_2, \dots, x_n\}$ be the set of input feature vectors and $Y = \{y_1, y_2, \dots, y_k\}$ the set of class labels. Then

$$X \times Y = \{(x, y) \mid x \in X, y \in Y\}$$

represents the set of all possible (input, label) pairings.

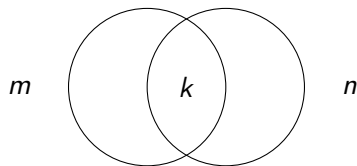
Inclusion–Exclusion Principle

Definition

A counting technique used to determine the number of elements in the union of two or more sets, by adding the sizes of the sets and subtracting the sizes of their intersections, to avoid double counting.

Two sets:

$$|A \cup B| = |A| + |B| - |A \cap B|$$



Three sets:

$$\begin{aligned} |A \cup B \cup C| &= |A| + |B| + |C| \\ &\quad - |A \cap B| - |A \cap C| - |B \cap C| \\ &\quad + |A \cap B \cap C| \end{aligned}$$

$$|A \cup B| = m + k + n + k - k = m + k + n$$

Inclusion–Exclusion: General Form & Example

General Form

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_i |A_i| - \sum_{i < j} |A_i \cap A_j| + \sum_{i < j < k} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} \left| \bigcap_{i=1}^n A_i \right|$$

Example: Market-Basket Transactions

Let T_1 = set of transactions containing Milk, T_2 = set of transactions containing Bread. The total number of transactions containing Milk *or* Bread is

$$|T_1 \cup T_2| = |T_1| + |T_2| - |T_1 \cap T_2|$$

Partially Ordered Set (Poset)

Definition

A relation R on a set A is a **partial order** if it is:

- **Reflexive** – aRa for all $a \in A$
- **Antisymmetric** – aRb and $bRa \implies a = b$
- **Transitive** – aRb and $bRc \implies aRc$

The set A together with R is called a **partially ordered set** (poset).

Example 1: $\leq$ on $A = \{1, 2, 3, 4, 5\}$

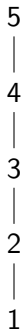
$$aRb \iff a \leq b$$

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (2, 2), (2, 3), (2, 4), (2, 5), \\ (3, 3), (3, 4), (3, 5), (4, 4), (4, 5), (5, 5)\}$$

Hasse Diagram: Chain $\{1, 2, 3, 4, 5\}$

Idea

A **Hasse diagram** is a simplified drawing of a poset: reflexive loops and transitive edges are omitted, and $a < b$ is drawn with b higher than a , connected by an edge only if there is no element strictly between them (covering relation).



Poset Example 2: Power Set of $\{1, 2, 3\}$ under $\subseteq$

Setup

$A = \{1, 2, 3\}$, $S = \mathcal{P}(A) = \{S_0, S_1, \dots, S_7\}$ where

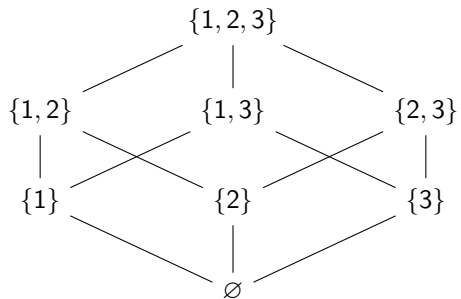
$S_0 = \emptyset$, $S_1 = \{1\}$, $S_2 = \{2\}$, $S_3 = \{3\}$, $S_4 = \{1, 2\}$, $S_5 = \{1, 3\}$, $S_6 = \{2, 3\}$, $S_7 = \{1, 2, 3\}$

Relation

$$S_i R S_j \iff S_i \subseteq S_j$$

This relation is reflexive, antisymmetric and transitive $\implies (\mathcal{P}(A), \subseteq)$ is a poset.

Hasse Diagram: $(\mathcal{P}(\{1, 2, 3\}), \subseteq)$



Thank You